Algebra 10 — Algebraic fractions

Calculator allowed. Marks in brackets. ★ = harder question. Factorise first, every time.

1 Simplify [4]

a $\dfrac{x^2 - 3x}{x - 3}$ **b** $\dfrac{x^2 - 4}{3x + 6}$ **c** $\dfrac{x^2 + 7x + 10}{x^2 + 3x - 10}$

2 Express as a single fraction in its simplest form [6]

a $\dfrac{3(x+2)}{4} - \dfrac{x-1}{6}$ **b** $\dfrac{x}{x^2 + x - 6} + \dfrac{2}{x + 3}$

3 Simplify [6]

a $\dfrac{4x}{2x + 8} \times \dfrac{x^2 + 6x + 8}{x^2}$ **b** $\dfrac{x^2 - 2x - 8}{x + 1} \div \dfrac{x + 2}{3x^2 + 3x}$

4 Solve [9]

a $\dfrac{2x + 1}{3} + x = 7$ **b** $\dfrac{3}{x} + \dfrac{x}{x + 2} = 2$ **c** $\dfrac{x + 2}{x - 1} - \dfrac{x + 1}{x + 3} = 1$

Total: 25 marks

Check: factorise top and bottom, then cancel whole brackets (never single terms) · add/subtract: common denominator = product of the factors, each used once · divide: flip the second fraction and multiply · equations: multiply every term by the lowest common denominator, then solve · substitute back to check.